CI3249 Advanced Data Mining

Project Brief

BasketIQ: Retail Market Intelligence Platform

Total: 20 marks

A. Project Description

Teams build one software product, BasketIQ, a market-intelligence system for a retail transaction dataset and perform relevant data mining tasks.

Dataset: Online Retail II, UCI Machine Learning Repository. Download from https://archive.ics.uci.edu/dataset/502/online+retail+ii .

B. Required Tasks

1. Perform exploratory analysis and produce a dimensionality-reduction visualisation of customers (PCA, t-SNE, or UMAP, whichever is appropriate).

2. Mine association rules from the baskets and evaluate them with support, confidence, and lift .

3. Segment customers with DBSCAN. Conduct a study with different parameter, compute silhouette scores, and interpret the noise points. Compare the segments with the k-means segments from the CI3237 batch layer.

4. Write a complete report on the above tasks, evaluation process, and results.

C. Deliverables

1. Source code or notebooks for the exploratory analysis, mining, and evaluation.

2. Report (1,500-2,000 words max). Do NOT exceed the Words limit.

3. Demonstration of the system with the CI3237 project (10-15 minutes, live).

4. Peer evaluation report.

5. Rubric

Criterion	MILO	Marks
Exploratory analysis and dimensionality-reduction visualisation	2	4
Association rule mining and evaluation (support, confidence, lift)	2, 3	5
DBSCAN segmentation and evaluation (parameter study, silhouette, noise interpretation, comparison with k-means)	2, 3	6
Report and Demo	5	5
Total		20